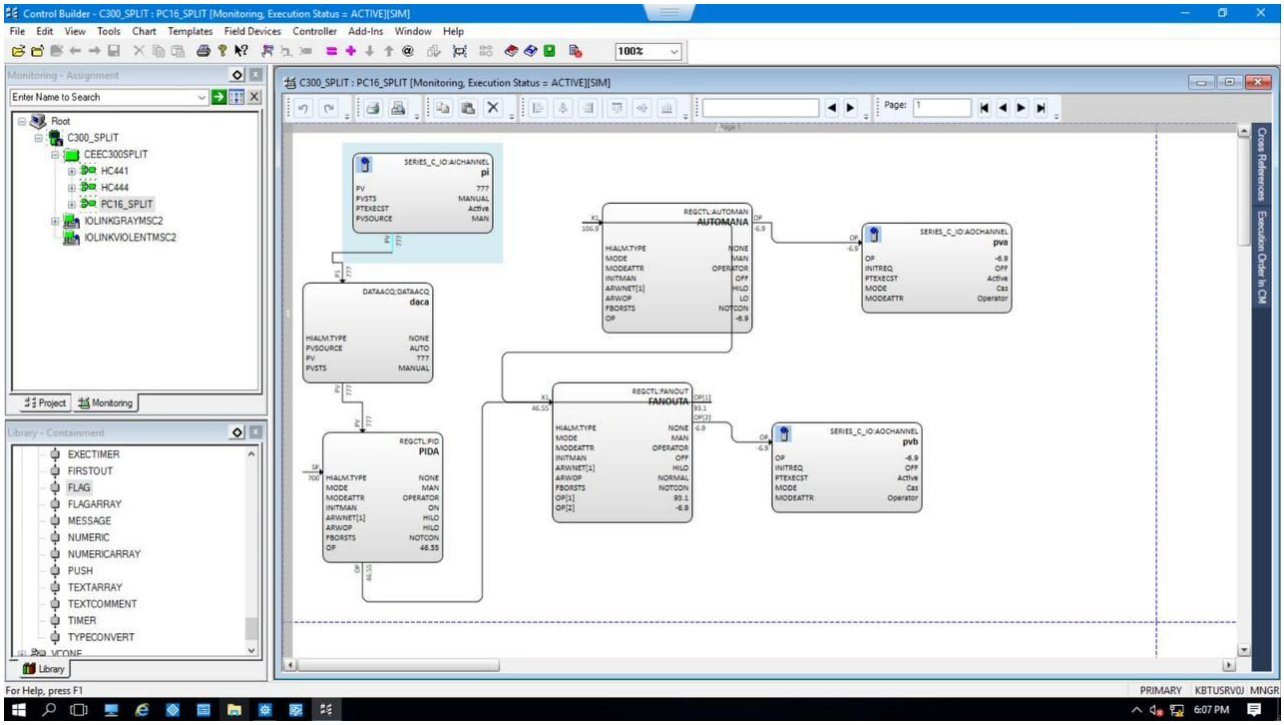
School of Information Technology and Engineering

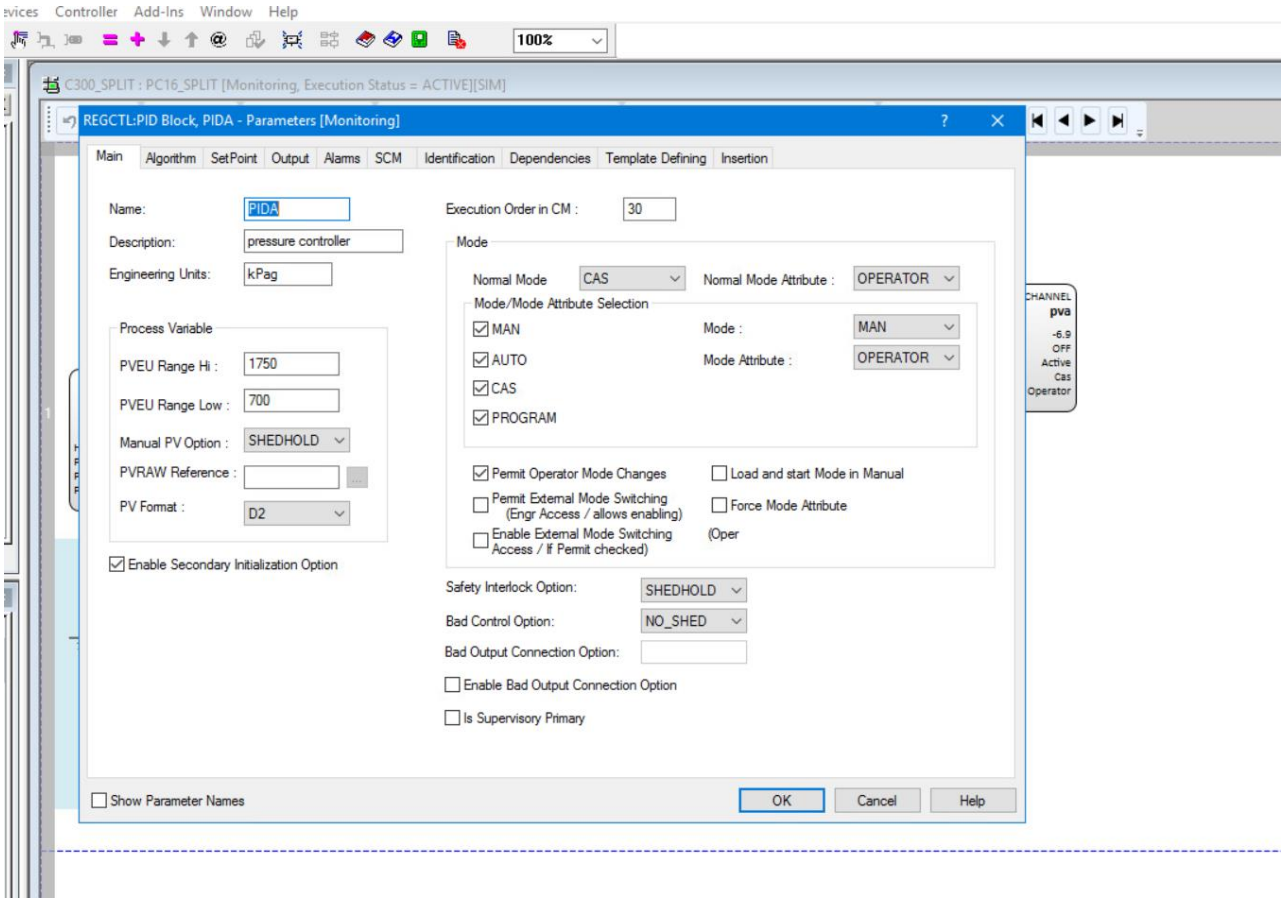
Advanced PLC programming

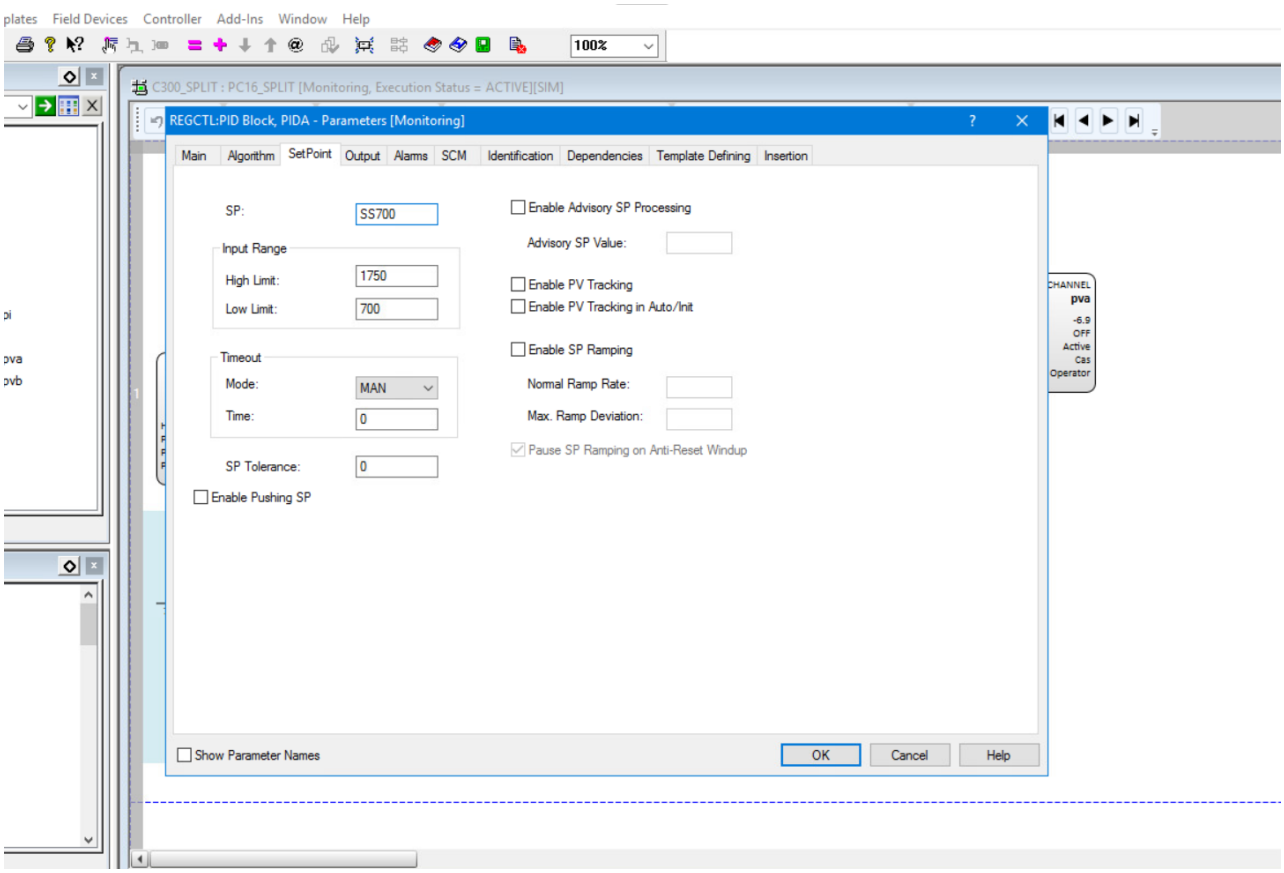
PID controller with Split Range control

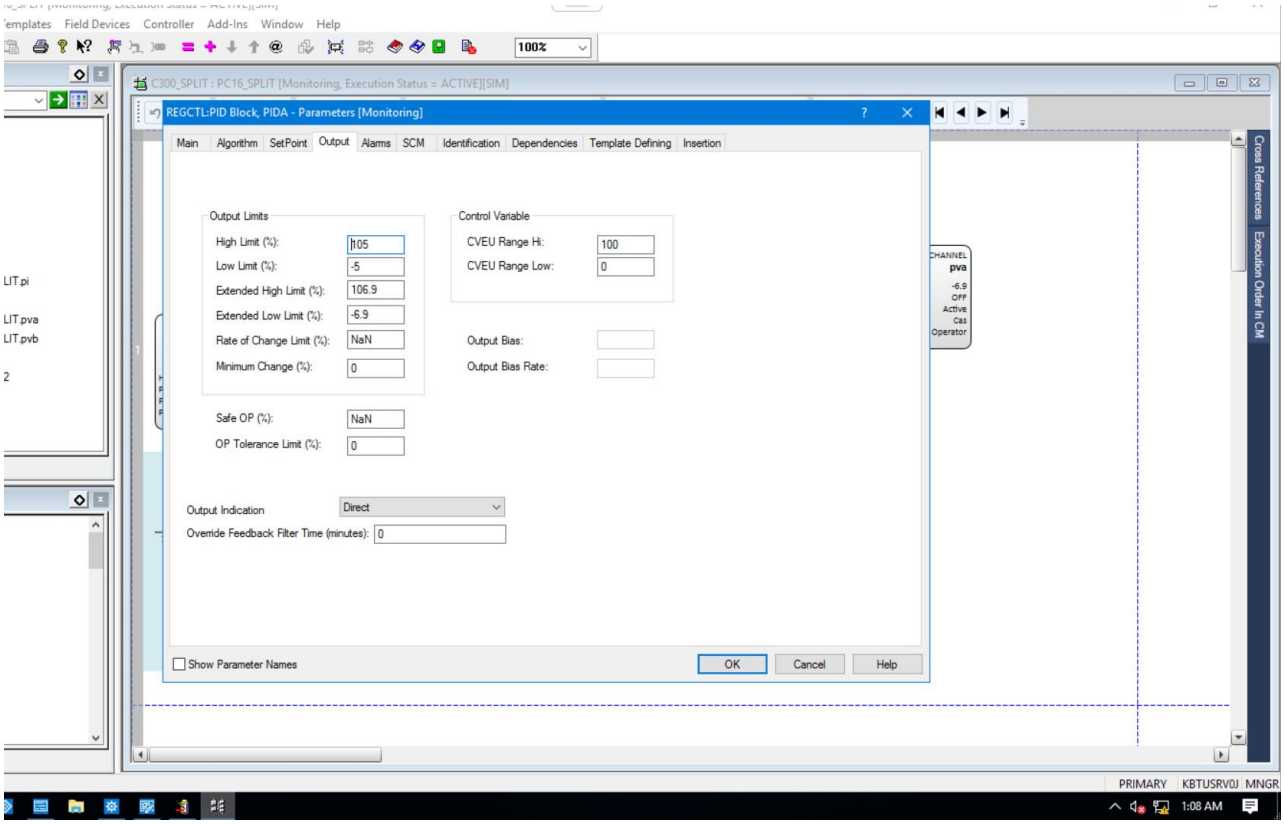
Done by: Saifolla Aknur

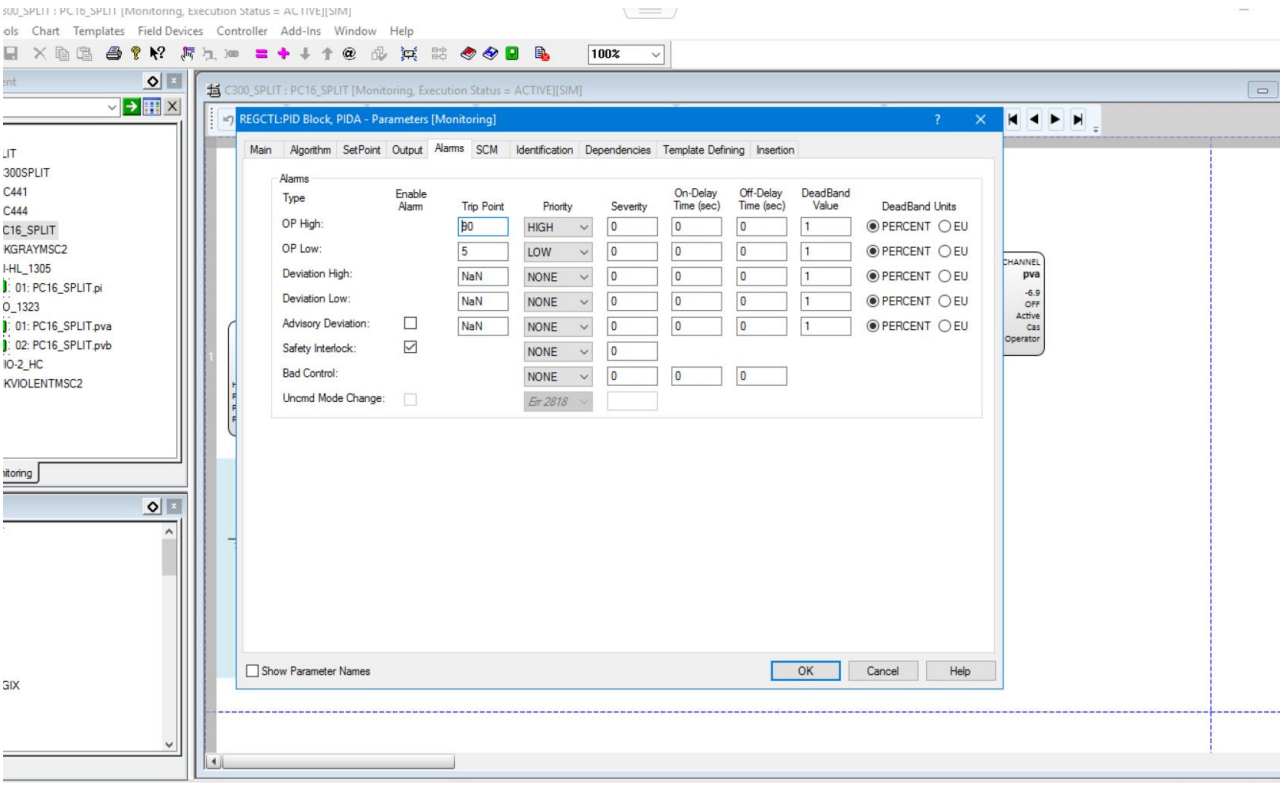
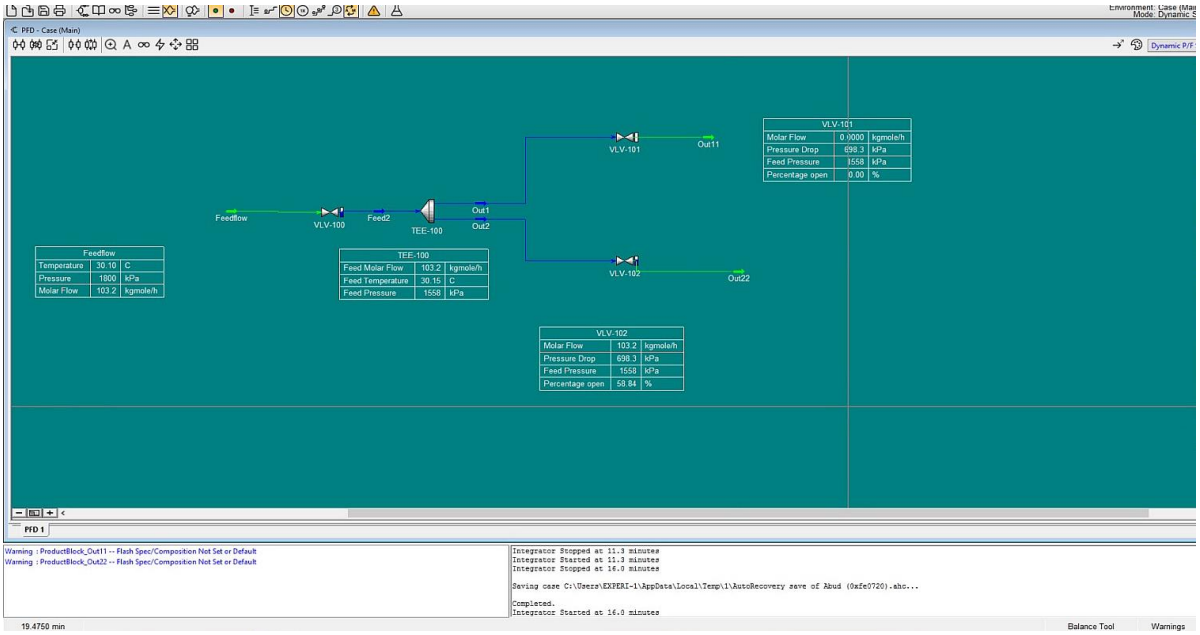
Almaty 2025


Control Builder - C300_SPLIT : PC16_SPLIT [Monitoring, Execution Status = ACTIVE][SIM]
File Edit View Tools Chart Templates Field Devices Controller Add-Ins Window Help
C300_SPLIT : PC16_SPLIT [Monitoring, Execution Status = ACTIVE][SIM]
Root
C300_SPLIT
CEEC300SPLIT
HC441
HC444
PC16_SPLIT
IOLINKGRAYMSC2
IOLINKVIOLENTMSC2
EXECTIMER
FIRSTOUT
FLAG
FLAGARRAY
MESSAGE
NUMERIC
NUMERICARRAY
PUSH
TEXTARRAY
TEXTCOMMENT
TIMER
TYPECONVERT
pi
daca
AUTOMANA
pva
FANOUTA
PIDA
pvb
For Help, press F1


C300_SPLIT : PC16_SPLIT [Monitoring, Execution Status = ACTIVE][SIM]
REGCTL:PID Block, PIDA - Parameters [Monitoring]
Main
Algorithm
SetPoint
Output
Alarms
SCM
Identification
Dependencies
Template Defining
Insertion
Name:
PIDA
Description:
pressure controller
Engineering Units:
kPag
Process Variable
PVEU Range Hi :
1750
PVEU Range Low :
700
Manual PV Option :
SHEDHOLD
PVRAW Reference :
PV Format :
D2
Enable Secondary Initialization Option
Execution Order in CM :
30
Mode
Normal Mode
CAS
Normal Mode Attribute :
OPERATOR
Mode/Mode Attribute Selection
MAN
AUTO
CAS
PROGRAM
Mode :
MAN
Mode Attribute :
OPERATOR
Permit Operator Mode Changes
Permit External Mode Switching (Engr Access / allows enabling)
Enable External Mode Switching Access / If Permit checked)
Load and start Mode in Manual
Force Mode Attribute
Safety Interlock Option:
SHEDHOLD
Bad Control Option:
NO_SHED
Bad Output Connection Option:
Enable Bad Output Connection Option
Is Supervisory Primary
Show Parameter Names
OK
Cancel
Help


plates Field Devices Controller Add-Ins Window Help
100%
C300_SPLIT : PC16_SPLIT [Monitoring, Execution Status = ACTIVE][SIM]
REGCTL:PID Block, PIDA - Parameters [Monitoring]
Main Algorithm SetPoint Output Alarms SCM Identification Dependencies Template Defining Insertion
SP: SS700
Input Range
High Limit: 1750
Low Limit: 700
Timeout
Mode: MAN
Time: 0
SP Tolerance: 0
Enable Pushing SP
Enable Advisory SP Processing
Advisory SP Value:
Enable PV Tracking
Enable PV Tracking in Auto/Init
Enable SP Ramping
Normal Ramp Rate:
Max. Ramp Deviation:
Pause SP Ramping on Anti-Reset Windup
Show Parameter Names
OK
Cancel
Help
CHANNEL
pva
-6.9
OFF
Active
Cas
Operator


Field Devices Controller Add-Ins Window Help
100%
C300_SPLIT : PC16_SPLIT [Monitoring, Execution Status = ACTIVE][SIM]
REGCTL:PID Block, PIDA - Parameters [Monitoring]
Main Algorithm SetPoint Output Alarms SCM Identification Dependencies Template Defining Insertion
Output Limits
High Limit (%): 105
Low Limit (%): -5
Extended High Limit (%): 106.9
Extended Low Limit (%): -6.9
Rate of Change Limit (%): NaN
Minimum Change (%): 0
Safe OP (%): NaN
OP Tolerance Limit (%): 0
Control Variable
CVEU Range Hi: 100
CVEU Range Low: 0
Output Bias:
Output Bias Rate:
Output Indication Direct
Override Feedback Filter Time (minutes): 0
Show Parameter Names
OK
Cancel
Help
Cross References
Execution Order In CM
PRIMARY
KBTUSRV0J MNGR
1:08 AM



Connect CB with UniSim